LINEAR INDEPENDENCE

The Wronskian Test

Nonzero at one point proves independence; zero everywhere proves nothing by itself.

THE THEOREM SAYS

If $W(f_1, \ldots, f_n)(x_0) \neq 0$ at any single point x_0, the functions are linearly independent.

IT DOES NOT SAY

~~If the Wronskian is 0 at a point, or even everywhere, the functions are dependent.~~

TESTING THE WRONSKIAN AT ONE POINT

WRONG WORKING

x and x^2 are dependent, because $W = x \cdot 2x - x^2 \cdot 1 = x^2$ is 0 at $x = 0$.

RIGHT WORKING

At $x = 1$, $W = 1 \neq 0$, so the functions are independent. One point with $W \neq 0$ is enough; one zero is not.

Converse with a condition: a Wronskian that vanishes identically implies dependence only when the functions solve one linear ODE with continuous coefficients.